

SQL Joins Exercises

INNER JOIN

Write a query to get a list of all students who are enrolled in at least one course along with the course names.

Tables Involved:

- Students
- Enrollments
- Courses

Expected Output:

- StudentID
- Student Name
- Course Name

Hint: Use an INNER JOIN between the Students table, the Enrollments table, and the Courses table.

LEFT JOIN

Write a query to get a list of all students, including those who are not enrolled in any courses. Show the student's name along with the course name. If a student is not enrolled in any course, the course name should be NULL.

Tables Involved:

- Students
- Enrollments
- Courses

Expected Output:

- StudentID
- Student Name
- Course Name (may be NULL for students not enrolled in any course)

Hint: Use a LEFT JOIN between the Students and Enrollments tables, and then join with the Courses table.

RIGHT JOIN

Write a query to get a list of all courses along with the students enrolled in them. Show the course name and the student's name. If a course has no students enrolled, the student's name should be NULL.

Tables Involved:

- Courses
- Enrollments
- Students

Expected Output:

- Course Name
- Student Name (may be NULL for courses with no enrolled students)

Hint: Use a RIGHT JOIN between the Courses and Enrollments tables, and then join with the Students table.

FULL OUTER JOIN

Write a query to get a list of all students and courses, showing all possible combinations. If a student is not enrolled in a course or a course has no students enrolled, it should still show the student or course with NULL values where applicable.

Tables Involved:

- Students
- Courses
- Enrollments

Expected Output:

- StudentID
- Student Name
- Course Name (may be NULL for students not enrolled in any course or for courses with no students enrolled)

Hint: Use a FULL OUTER JOIN between the Students and Courses tables, and include the Enrollments table.

JOIN with Multiple Tables

Write a query to get a list of all students along with their grades, course names, and instructors. Show the student's name, the course name, the grade, and the instructor's name.

Tables Involved:

- Students
- Enrollments
- Courses
- Instructors

Expected Output:

- Student Name
- Course Name
- Grade
- Instructor Name

Hint: Use INNER JOIN to join the Students, Enrollments, Courses, and Instructors tables.

JOIN with Aggregate Function

Write a query to get the average grade of each student. Show the student's name and the average grade for all courses they are enrolled in.

Tables Involved:

- Students
- Enrollments

Expected Output:

- Student Name
- Average Grade

Hint: Use INNER JOIN between the Students and Enrollments tables and use the AVG() aggregate function to calculate the average grade for each student.

JOIN with WHERE Condition

Write a query to get a list of all students who are enrolled in a course with the instructor 'Dr. Alice Johnson'. Show the student's name and the course name.

Tables Involved:

- Students
- Enrollments
- Courses
- Instructors

Expected Output:

- Student Name

- Course Name

Hint: Use INNER JOIN between Students, Enrollments, Courses, and Instructors, and apply a WHERE condition to filter by the instructor's name.